

Pseudonymized but Linkable: Cross-Document Re-identification Risks in LLM-Ready Sensitive Text Corpora

Anonymous authors

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Abstract

1 Sensitive text corpora are increasingly prepared for retrieval, evaluation,
2 sharing, and model adaptation by transforming one document at a time.
3 This document-local workflow removes obvious direct identifiers, but it
4 can leave a corpus-level failure mode: records with no names, emails, or ac-
5 count numbers may still be linkable through repeated quasi-identifiers such
6 as role, region, institution type, family context, health context, financial
7 context, schedules, and rare events. We introduce CrossDoc-PrivacyBench,
8 a fully synthetic benchmark with 120 personas, 480 documents, risk tiers,
9 utility labels, decoy auxiliary profiles, and RAG-style exposure tasks. We
10 show that common document-local transformations remain linkable: direct
11 redaction leaves local auxiliary matching at 0.71 top-1, a Presidio-style
12 PII baseline remains at 0.59, and a GPT-5.5 auxiliary attacker matches
13 direct-redacted and document-local corpora almost perfectly on a 48-person
14 T2/T3 stress audit. We propose LinkGuard, a simple corpus-aware general-
15 ization heuristic that estimates cross-document uniqueness and generalizes
16 high-risk quasi-identifiers until a target anonymity set is reached. At target
17 $k = 5$, LinkGuard reduces local auxiliary top-1 to 0.04 while preserving
18 issue classification and retrieval Recall@5 at 1.00; aggressive redaction
19 reaches similar local privacy but drops issue accuracy to 0.35 and retrieval
20 Recall@5 to 0.31. These synthetic results suggest that LLM-ready data trans-
21 formations should be evaluated for corpus-level linkage resistance, not only
22 document-local PII removal.

1 Introduction

24 Sensitive data is often most useful when it is not isolated. A hospital may have clinical
25 messages, billing notes, scheduling records, and accommodation requests about the same
26 patient. A legal-aid organization may have intake notes, benefits appeals, landlord disputes,
27 and follow-up emails about the same client. A company may have support tickets, HR
28 notes, finance requests, and internal chat snippets about the same employee or customer.
29 Modern foundation-model workflows, retrieval-augmented generation (RAG), evaluation-
30 set construction, model adaptation, analytics, and data sharing, therefore often operate over
31 *multi-document* corpora rather than independent documents (Lewis et al., 2020; Zeng et al.,
32 2024; Jiang et al., 2024).

33 The dominant operational response is document-local de-identification: remove names,
34 emails, phone numbers, mailing addresses, account numbers, and other spans that look
35 identifying within one record. That response is necessary, but it misses the central lesson of
36 quasi-identifier privacy: combinations of ordinary attributes can distinguish people even
37 when direct identifiers are gone (Sweeney, 2002; Narayanan and Shmatikov, 2008; Cohen,
38 2022). In text, this problem is harder because the identifiers are semantic rather than tabular.
39 A region, occupation, institution type, care schedule, family constraint, legal context, and
40 local event may each look harmless in isolation, yet together they can form a stable signature
41 across documents. Pseudonymization can make the problem worse: a stable pseudonym
42 helps preserve continuity for benign users, but it is also a cross-document linkage handle.

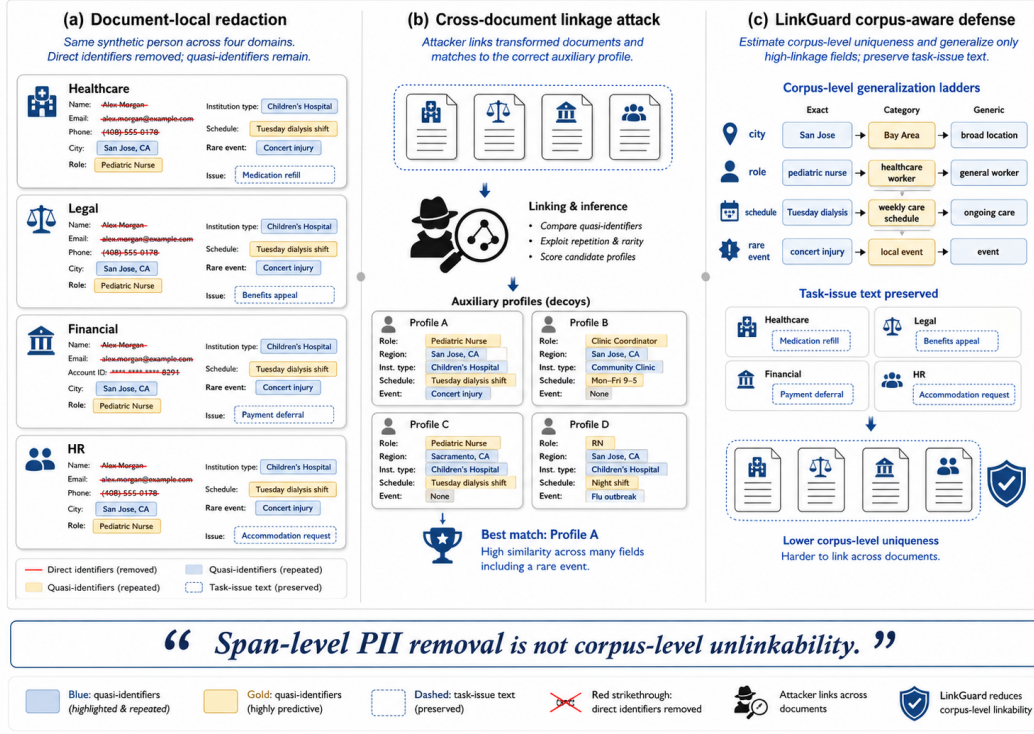


Figure 1: Conceptual overview of the cross-document linkage failure mode and the corpus-aware generalization defense.

LLM-ready data makes this failure mode more urgent. Recent work shows that language models can infer personal attributes from text even when the text does not contain explicit identifiers (Staab et al., 2024); text anonymization benchmarks increasingly emphasize residual re-identification risk rather than span recall alone (Pilán et al., 2022; Krčo et al., 2026); and RAG privacy work shows that retrieval databases can leak private records through retrieval and generation channels (Zeng et al., 2024; Jiang et al., 2024). This paper studies a focused question: *after document-local anonymization or pseudonymization, can an attacker still link multiple transformed documents to the same person and match them to auxiliary background profiles?* We avoid real-person data and real re-identification attempts. Instead, every persona, document, decoy, and utility label is generated synthetically, giving us safe ground truth for controlled privacy auditing.

Our contributions are:

- **Benchmark.** CrossDoc-PrivacyBench is a synthetic cross-document privacy benchmark with 120 personas, 480 documents, risk tiers, utility labels, document templates, and decoy auxiliary profiles.
- **Audit protocol.** We evaluate transformations with document linkage, auxiliary-profile matching, quasi-identifier recovery, RAG-style exposure, GPT-5.5 stress audits, and utility retention.
- **Finding.** Direct redaction, Presidio-style PII removal, document-local anonymization, and stable pseudonyms all leave strong residual linkage risk.
- **Defense.** LinkGuard is a simple corpus-aware generalization heuristic that improves the synthetic privacy-utility tradeoff over blanket aggressive redaction.

2 Threat Model and Benchmark

Setting. A data holder has a sensitive text corpus with multiple records per person, household, customer, patient, client, or employee. The corpus is transformed before it is indexed for RAG, shared internally, released for evaluation, or used in model-adaptation workflows. The holder may know which records belong to the same account, case, patient, or synthetic persona; our method uses this grouping only to estimate repeated quasi-identifier risk.

Attacker. The attacker sees the transformed corpus. For the auxiliary-profile task, the attacker also receives a small candidate set containing one true synthetic auxiliary profile and decoys. This simulates background knowledge such as a directory, leaked metadata table, public biographical context, or another internal database. The attacker does not receive original direct identifiers, and candidate labels are neutralized in to avoid leaking synthetic persona IDs.

Benchmark. CrossDoc-PrivacyBench contains 120 synthetic personas and 480 documents. Each persona has direct identifiers, utility labels, and quasi-identifiers: region, city, occupation, employer type, education, family structure, medical context, financial context, legal context, schedule pattern, community affiliation, and rare event. Each persona has four short service records spanning healthcare, legal, financial support, and HR. Personas are assigned to three risk tiers: T1 contains mostly generic context, T2 contains moderately specific context, and T3 contains high-linkage combinations. For each persona, we build a ten-candidate auxiliary set with one true profile and nine decoys that share partial attributes. This design lets us measure whether a transformation protects against both document linkage and background-knowledge matching while retaining task utility.

3 Transformations and LinkGuard

We compare common document-local transformations against a corpus-aware alternative. C1 directly redacts names, email addresses, phone numbers, mailing addresses, and account IDs. C1b applies a Presidio-style PII analyzer plus local regex cleanup. C2 uses stable pseudonyms across the corpus, while C3 resets pseudonyms per document. C4 is a document-local anonymization proxy that generalizes selected local spans without seeing neighboring documents. C5 is LinkGuard. C6 is aggressive redaction, suppressing most quasi-identifiers and some task issue phrases. We also run GPT-5.5 document-local anonymization as an stress baseline on a synthetic subset.

Key idea. Document-local anonymizers ask, “Is this span identifying in this document?” LinkGuard asks, “Does this repeated detail make the person distinguishable in the corpus?” For each quasi-identifier field, LinkGuard defines a three-level ladder:

exact value → semantic category → generic suppression.

For example, a city becomes a region and then “a broad location”; an occupation becomes a job family and then “general worker”; a rare event becomes a local-event category and then “a local event.”

Algorithm. For each persona group, LinkGuard starts from direct redaction and computes the current estimated anonymity set: the number of personas sharing the same generalized quasi-identifier representation. If this estimate is below the target k , the method chooses another field to generalize. The field score balances value rarity, a specificity prior, and an approximate utility cost:

$$s(f) = \frac{\text{rarity}(f) + \text{specificity}(f) + 1}{\max(\text{utility_cost}(f), 0.1)}.$$

Fields with rarer values, higher specificity, and lower expected utility cost are generalized first. The process stops when the estimated anonymity set reaches target k or all fields have

Cond.	Transformation	Local Aux@1	GPT Aux@1	Issue	Ret@5
C1	direct PII redaction	0.71	1.00	1.00	1.00
C1b	Presidio-style PII	0.59	0.98	1.00	1.00
C4	document-local anon.	0.60	1.00	1.00	1.00
C5	LinkGuard	0.04	0.44	1.00	1.00
C6	aggressive redaction	0.05	0.33	0.35	0.31

Table 1: Core privacy–utility summary. Local Aux@1 is measured on 96 held-out personas using the deterministic local attacker. GPT Aux@1 is the 48-person T2/T3 GPT-5.5 auxiliary-matching stress audit. Issue and Ret@5 are local utility metrics. Lower Aux@1 is better for privacy; higher Issue and Ret@5 are better for utility.

102 been generalized or suppressed. This is a heuristic risk-budgeting procedure, not a formal
 103 privacy guarantee: its estimate depends on the corpus, schema, grouping assumptions, and
 104 attacker knowledge.

105 4 Attacks and Metrics

106 **Linkage and auxiliary matching.** We evaluate same-person pairwise linkage, fixed- K
 107 clustering, and auxiliary-profile matching. The main local auxiliary attacker uses word TF-
 108 IDF; robustness checks use character TF-IDF, hybrid TF-IDF, and a field-weighted matcher
 109 that explicitly scans exact and generalized schema fields. We report top-1 and top-3 auxiliary
 110 accuracy and mean reciprocal rank; lower is better for privacy.

111 **Quasi-identifier recovery.** Auxiliary matching is an end-to-end privacy signal, but it does
 112 not explain what leaked. We therefore scan transformed documents for exact and coarse
 113 recovery of schema fields such as city, role, institution type, family context, medical context,
 114 legal context, schedule, affiliation, and rare event.

115 **RAG exposure.** We issue profile-like queries against each transformed corpus and measure
 116 whether target-persona documents appear in the top retrieved results. We also run a small
 117 GPT-5.5 RAG-generation audit in which retrieved T3 contexts are passed to an instruction-
 118 following model to test whether it can infer that the documents describe the same synthetic
 119 person.

120 **Utility.** Utility is measured with issue classification, task-oriented retrieval Recall@5, and
 121 lightweight fact preservation. These metrics test whether service-triage information survives
 122 transformation.

123 5 Results

124 **Summary.** The main result is simple: document-local transformations preserve cross-
 125 document signatures, while corpus-aware generalization removes most local linkage signal
 126 without the utility loss of blanket redaction. Table 1 reports the compact version of the story.
 127 Detailed robustness numbers are easier to compare visually; Figures 2–4 summarize the
 128 privacy–utility frontier, corpus-level mechanism, and RAG stress audits.

129 **Document-local transformations remain linkable.** Direct redaction preserves the quasi-
 130 identifier signature: the local auxiliary attacker matches the correct profile as often as on
 131 the original corpus. The Presidio-style baseline removes more obvious spans, but still
 132 leaves high residual matching. The document-local anonymization proxy and GPT-5.5
 133 document-local anonymizer show the same weakness: because each document is trans-
 134 formed independently, repeated role, location, schedule, institution, and rare-event patterns
 135 remain available across the corpus. Stable pseudonyms create a different failure mode: the
 136 pseudonym itself becomes a linkage handle, making pairwise linkage nearly trivial.

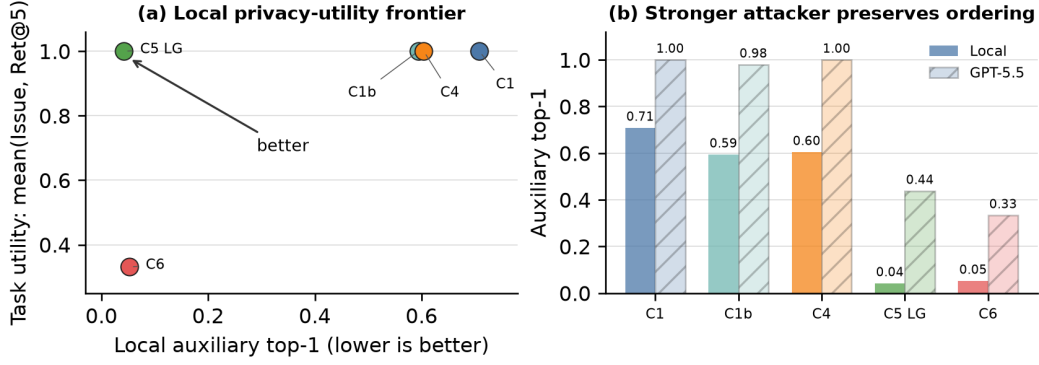


Figure 2: Privacy-utility frontier and stronger-attacker stress audit. Left: local auxiliary top-1 versus task utility. Right: deterministic local matching and GPT-5.5 auxiliary matching preserve the same qualitative ordering.

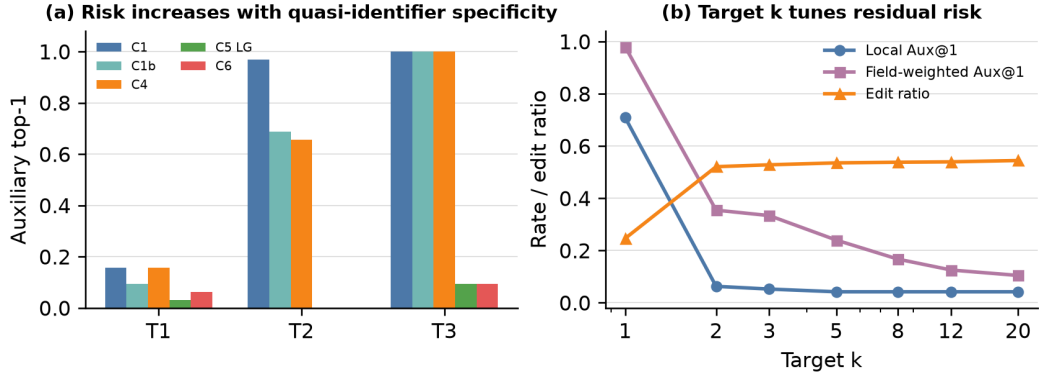


Figure 3: Corpus-level linkage mechanism. Left: residual auxiliary matching rises with quasi-identifier specificity under document-local transformations. Right: LinkGuard gives a tunable target- k risk budget that trades residual linkage against edit distance.

137 **LinkGuard improves the privacy-utility frontier.** LinkGuard sharply reduces local auxiliary
 138 matching and exact quasi-identifier recovery while preserving issue classification and
 139 retrieval utility. Aggressive redaction reaches a similar local privacy level, but it removes
 140 task-relevant issue information and hurts retrieval. This contrast is the main practical lesson:
 141 for LLM-ready corpora, the best transformation is not always the most destructive one; it is
 142 the one that removes the high-linkage fields while retaining task information.

143 **The risk is corpus-level, not just span-level.** Risk-tier analysis confirms the benchmark
 144 behaves as intended: direct redaction is mostly adequate on generic T1 personas, but
 145 fails on T2 and T3 personas where repeated quasi-identifiers are more specific. Ablations
 146 show that role/institution and location are the largest single drivers, while family, health,
 147 schedule, and rare-event fields become important in combination or under a field-aware
 148 attacker. A corpus-awareness ablation further supports the method design: using true
 149 corpus co-occurrence statistics reaches the target anonymity coverage, while shuffled co-
 150 occurrence planning and global one-level generalization either miss the target or leave
 151 strong field-aware matching.

152 **GPT-5.5 and RAG audits corroborate the local story.** The GPT-5.5 auxiliary audit is in-
 153 tententionally a time-stamped stress test rather than the main reproducible evidence, but it
 154 strengthens the conclusion. On the T2/T3 subset, the model almost perfectly matches direct-
 155 redacted, Presidio-redacted, and document-local corpora; LinkGuard remains substan-

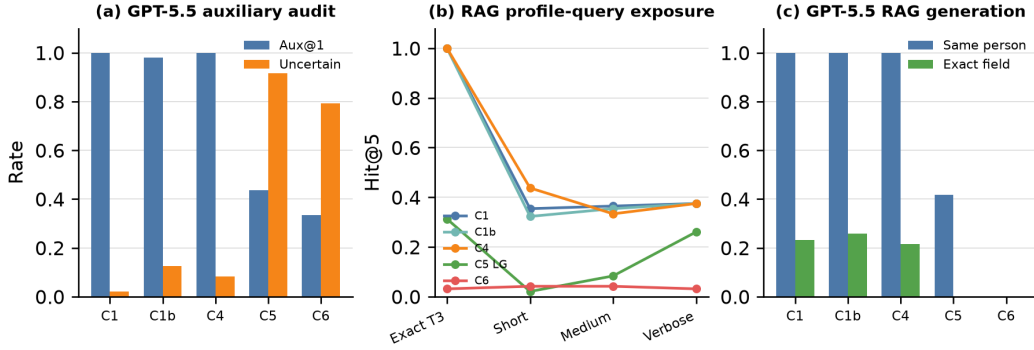


Figure 4: RAG stress audits. GPT-5.5 auxiliary matching, profile-query retrieval, and RAG generation preserve the local ordering: LinkGuard reduces corpus-level linkage risk, but does not provide a formal guarantee.

156 tially less matchable and much more uncertain, though not risk-free. A GPT-5.5 evidence-
 157 extraction audit explains why: successful direct-redaction matches cite location and role,
 158 while LinkGuard residual matches rarely cite high-specificity fields and are usually marked
 159 uncertain. The RAG diagnostics tell the same story from a retrieval perspective: exact and
 160 generated profile-like queries easily retrieve high-linkage direct/document-local records,
 161 whereas LinkGuard greatly reduces exact field recovery in retrieved contexts.

162 **Robustness.** The qualitative ordering holds across three synthetic corpus seeds, a noisy-
 163 style rerendering of the documents, auxiliary candidate sets of size 10/20/50, and
 164 word/character/hybrid/field-weighted local attackers. The most important caveat is that
 165 the field-weighted attacker still finds residual coarse-context signal under LinkGuard. This
 166 is expected: LinkGuard generalizes high-risk fields; it does not erase all semantics.

167 6 Related Work

168 Classic quasi-identifier privacy work motivates our setting. k -anonymity introduced gen-
 169 eralization and suppression as ways to reduce record uniqueness (Sweeney, 2002), while
 170 later work showed that sparse or high-dimensional records can be de-anonymized with
 171 limited auxiliary knowledge (Narayanan and Shmatikov, 2008). Subsequent attacks and
 172 critiques show that quasi-identifier de-identification can fail even when many attributes are
 173 treated as potential identifiers (Cohen, 2022). Related privacy models such as ℓ -diversity
 174 and t -closeness address sensitive-attribute homogeneity and distributional leakage, but they
 175 remain difficult to apply directly to free text (Machanavajjhala et al., 2007; Li et al., 2007).

176 Text anonymization benchmarks increasingly move beyond named-entity deletion. TAB pro-
 177 vides legal-text annotations and privacy-oriented evaluation for text anonymization (Pillán
 178 et al., 2022). RAT-Bench explicitly evaluates residual re-identification risk from direct and
 179 indirect identifiers in text, which is closely aligned with our emphasis on quasi-identifiers
 180 (Krčo et al., 2026). Surveys of anonymization techniques and metrics emphasize that pri-
 181 vacy, utility, legal requirements, and human expectations are often measured inconsistently
 182 (Deußner et al., 2025; Ren et al., 2025). Tools such as Presidio and OpenAI Privacy Filter are
 183 useful PII-detection components, but PII detection alone is not a complete anonymization
 184 guarantee (Microsoft, 2026; OpenAI, 2026).

185 LLM privacy work motivates stronger attackers. Language models can infer private at-
 186 tributes from text beyond direct memorization (Staab et al., 2024), and large models can
 187 reveal memorized training data under extraction attacks (Carlini et al., 2021; Nasr et al.,
 188 2023). RAG systems add a retrieval database that can itself leak sensitive records (Zeng et al.,
 189 2024; Jiang et al., 2024). Our contribution is complementary: we focus on cross-document

linkage after document-level transformations, where repeated corpus-level quasi-identifiers can survive otherwise reasonable LLM-ready anonymization workflows.

7 Limitations and Ethics

The benchmark is synthetic and template-controlled. This gives safe ground truth and reproducibility, but limits external validity. Real sensitive corpora may contain richer writing styles, temporal dependencies, institutional terminology, missing fields, contradictions, OCR errors, and adversary knowledge.

LinkGuard is a heuristic generalization method, not a formal privacy guarantee. Utility is measured with issue labels, retrieval, and lightweight fact checks rather than full downstream deployments.

The benchmark is intended for defensive auditing. We do not collect, process, or attempt to re-identify real people. All examples are synthetic and generated from a controlled schema. Any extension to real data would require authorization, data minimization, access controls, logging, and ethical review. Practitioners should not treat success on this benchmark as proof that a real corpus is safe to release, index, or train on.

8 Conclusion

Document-local PII removal is not enough for LLM-ready sensitive corpora. In a multi-document setting, repeated quasi-identifiers can support linkage, auxiliary-profile matching, and profile-query retrieval even after direct identifiers are removed. Our synthetic benchmark suggests that privacy evaluation should include corpus-level linkage resistance, not only span-level PII recall. It also suggests that simple corpus-aware generalization can improve the privacy-utility frontier relative to both document-local anonymization and blanket redaction. The broader lesson is that data transformation for foundation models should ask what a capable attacker can still infer across the corpus, not only what identifying spans were removed from each document.

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